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Supporting Master's Students and Supervisors with Metacheck

Project Report

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1 Motivation and Procedure

The aim of this project was to extend Metacheck, an existing tool for automatically checking research outputs with modules that could be useful to master's students and their supervisors. Metacheck is a modular tool, meaning that each automated check is implemented as a separate module. It is available as an **R package** and a **Shiny app**. One of the motivations behind Metacheck was that students often make recurring mistakes in areas such as adherence to reporting guidelines, interpretation of p -values or documentation of research practices (e.g., the reporting of a power analysis). Identifying such issues can require substantial time from supervisors, particularly when they need to provide similar feedback repeatedly. Automated checks could therefore help students in identifying and addressing potential issues before submitting their thesis, while simultaneously reducing the burden on supervisors.

By now, Metacheck has developed into an extensible collection of automated checks for research outputs. Inspired by software development, where checks are used to identify potential problems before software is released, Metacheck screens research outputs (e.g., manuscripts and theses) for potential issues and provides information that can help researchers or students follow best practices. Importantly, these checks are intended to support rather than replace human expertise. For example, a check may identify a misreported statistical test or flag a misinterpreted p -value, but interpreting the result and deciding whether action is necessary remains the responsibility of the researcher or supervisor.

To better understand which aspects of thesis supervision could benefit from automated checks and how these align with Metacheck's current modules, we conducted a survey among scientific staff involved in supervising master's theses in the Department of Industrial Engineering & Innovation Sciences (IE&IS) at TU/e. The results of this survey are presented in the following section.

2 Survey of IE&IS Supervisors

The survey was shared in November 2025 with all IE&IS staff members involved in the supervision of master's theses, with 22 participating. The survey consisted of five open-ended questions addressing (1) the most common types of feedback supervisors provide to master's students on their theses, (2) which aspects of this feedback they believe could be (partially) automated, (3) which aspects they believe should not be automated, (4) whether they or their students follow specific reporting guidelines, such as APA, and (5) any other feedback they wished to share. The full survey can be found in [Appendix A](#)

2.1 Survey Results

The survey results indicate that supervisors provide feedback on a broad range of aspects of thesis quality, including both formal presentation and substantive content. Commonly mentioned issues included the coherence and narrative flow of the thesis, the connection between literature and the current study, the formulation and derivation of research questions and hypotheses, and the justification of methodological choices. Supervisors also frequently commented on formatting and structure, the quality of literature summaries and discussions, and citation and reference issues. Statistical reporting was another recurring area of feedback, including checking test assumptions, conducting and reporting power analyses, and providing complete information such as test statistics, degrees of freedom, and exact p -values.

Most respondents agreed that at least some of these aspects could be (partially) automated. Formatting, structure and layout were considered particularly suitable, as were checks of research questions, citations, and references. These aspects can generally be assessed against relatively explicit criteria or conventions. However, respondents also identified several areas that they considered less suitable for automation. These mainly involved tasks requiring content expertise, contextual understanding, or critical thinking, such as evaluating methodological choices, developing research questions, and assessing the substantive quality of a thesis. Opinions differed somewhat regarding the overall “storyline” of a thesis: while some considered this inherently dependent on human judgment, others suggested that large language models (LLMs) could potentially help improve the structure and clarity of a thesis. Respondents also emphasized that feedback should be tailored to the individual student and that automated tools should flag potential issues rather than evaluate or grade the overall quality of a thesis.

Some respondents questioned the desirability of automated feedback, particularly when using LLMs, and raised concerns about privacy and the costs of developing and maintaining such systems. They emphasized that any automated system should therefore be carefully tested before implementation.

Finally, most respondents reported using APA guidelines, while a small number mentioned Harvard or guidelines specific to particular conferences or journals. Overall, the results suggest that automation is most promising for checks based on explicit rules or conventions, whereas tasks requiring contextual judgment and content expertise should remain primarily the responsibility of students and supervisors.

3 Using Metacheck

The survey results suggest that automated checks could complement the existing supervision process. In particular, when supervisors repeatedly provide similar feedback on statistical reporting, citations and references, and the documentation of research practices, automated checks could allow students to detect and address some of these issues independently.

Metacheck is well suited to this role because its checks are designed as supporting tools rather than automated evaluations. The output of a module does not constitute a judgment of the overall quality of a thesis. Instead, it highlights a potential issue, provides guidance, and leaves the final interpretation and decision to the user.

Broadly speaking, Metacheck modules can be classified into four categories: **general modules**, **method modules**, **results modules** and **reference modules**. For a detailed description of all modules, readers are referred to the [Metacheck documentation](#). First, the general modules check for statements related to Open Science practices, including open and code sharing, conflict of interest, ethics and funding.

Second, the method modules check for preregistrations and the presence and reporting of power analyses. That is, if a power analysis is detected, the corresponding module provides feedback on whether the analysis is fully reported.

The results modules currently provide a broad collection of checks covering adherence to reporting guidelines, including the reporting and recomputation of effect sizes and p -values, interpretation of p -values, and best research practices when sharing data and code. Where research outputs are shared in repositories, it can additionally examine the organisation and documentation of data and code, identify potential data-quality and reproducibility problems, and check aspects of compliance with standards for organizing research data.

Finally, the reference modules can identify inconsistencies in reference information and cross-reference cited literature with databases containing information about retractions, replication attempts, and post-publication peer review (i.e., comments on PubPeer). These checks can be useful to identify potentially problematic references, such as papers for which replication attempts exist but are not cited, or papers with PubPeer comments, which may concern methodological issues.

The current functionality therefore aligns particularly well with the types of feedback that supervisors identified as potentially automatable: checks based on explicit reporting conventions, common statistical practices, citation and reference information, and other relatively well-defined criteria. At the same time, the survey highlighted several areas that are less suitable for automation, such as evaluating the overall quality of a thesis, judging methodological choices, assessing the substantive contribution of the research, or providing highly context-dependent feedback on its storyline. Metacheck does not attempt to replace this type of supervision. Instead, its role is to identify potential issues that can be checked systematically, allowing students to address more routine problems themselves and enabling supervisors to focus their time on feedback that requires expertise, context, and critical judgment.

Metacheck can apply multiple modules to a single research output and combine their results into a single report, allowing a thesis and its associated research outputs to be screened across several dimensions at once. This makes it possible to use Metacheck as a first layer of quality control before a supervisor reviews the work in detail. Metacheck can also apply the same set of modules to multiple research outputs simultaneously, allowing supervisors to screen several theses in a single analysis.

Finally, Metacheck does not use LLMs by default, but some modules can optionally use LLMs to extract and classify information from text. This can be useful for tasks like extracting information about

power analyses, whose reporting tends to vary considerably due to a lack of established reporting guidelines.

4 Applying Metacheck to Master's Theses

To illustrate how Metacheck can be applied to master's theses, we applied selected modules from the R package `metacheck` to 10 master's theses. We first searched the [TU/e library portal](#) on 10 August 2026 for master's theses completed in 2025 within the Human-Technology Interaction (HTI) group, which yielded 55 results. We then used the `sample()` function in R to randomly select 10 unique integers between 1 and 55, each corresponding to one master's thesis. The selected theses were subsequently manually downloaded as PDFs from the TU/e library portal.

The report was written in Quarto using R, including the `metacheck`, `tidyverse` and `kableExtra` packages. All materials, generated data and code required to reproduce this report can be found in the [project's Github repository](#).

Tip

Although the analyses in this report were conducted the R `metacheck` package, the most convenient way for supervisors and students to use Metacheck might be through its [Shiny app](#) which is hosted on the TU/e servers. Through it, users can run all Metacheck modules at once and generates a user-friendly [report](#) summarizing the results.

4.1 `ethics_check` module

The `ethics_check` module searches a paper for a statement indicating that the research received **ethics approval**, for example from an Institutional Review Board (IRB), a Research Ethics Committee, or an animal-care committee, or that the research was explicitly exempted from or waived of ethics approval. Reporting ethics approval is generally expected for studies involving human or animal participants.

To illustrate this, we use Thesis 1 and apply the module to identify sentences that may contain an ethics statement. If the module detects an ethics approval statement in a thesis that appears to involve data collection, it returns the following message:

Ethics Approval Check: All papers that appeared to involve live data collection contained an ethics approval statement.

Table 4.1 shows the three sentences identified by the module that might refer to an ethics statement. In the first case, the detected sentence is indeed an ethics statement.

Table 4.1: Potential ethics statements detected by the `ethics_check` module

text
The study was approved by the TU/e Ethical Review Board (experiment ID 2207), and all participants provided informed consent and received monetary compensation for their participation. After logging in, participants submitted the informed consent form (approved by the TU/e Ethical Review Board) and gave explicit consent to their participation and optionally consented to publicly sharing their recorded data in the Open Science Framework (n.d.).

Experiment Procedure Flowchart Next, participants were shown the informed consent form, which had been approved by the TU/e Ethics Review Board.

4.2 stat_p_nonsig module

The `stat_p_nonsig` module finds every non-significant p -value ($p \geq .05$) so a reader can check whether the surrounding text interprets it correctly. A common error is to treat a non-significant result as evidence that there is no effect (“there was no difference between groups, $p = .27$ ”), when a non-significant result is simply inconclusive.

Table 4.2 shows the first five non-significant p -values detected by the module, along with the sentence in which each p -value appears. This allows users to assess whether the p -value may have been misinterpreted. In this case, none of the sentences include a statement indicating that there was no difference.

Table 4.2: Non-significant p -values detected by the `stat_p_nonsig` module

text	expanded
$p = .924$	The test showed that individual differences ($F(31, 62) = 10.37, p < .001$) was a significant predictor while XAI condition ($F(2, 62) = 0.08, p = .924$) was not.
$p = .421$	No significant effect was found for condition as predictor, $\chi^2(1) = 0.649, p = .421$ (corrected for ties).
$p = .088$	The overall model was almost significant with Wald $\chi^2(12) = 19.01, p = .088$.
$p = 0.1033$	■ CS Correlation: $r = 0.982, p = 0.0184$ ■ Gold Correlation: $r = 0.897, p = 0.1033$
$p = 0.1033$	Gold accumulation also showed a strong positive correlation ($r = 0.897$), though the result was not statistically significant ($p = 0.1033$), likely due to the small sample size ($n = 4$ matchups) and the limitations of the gold estimation model.

4.3 marginal module

The `marginal` module lists sentences describing an effect as “marginally significant” (or a related phrase). Describing a non-significant result as “marginally significant”, “approaching significance”, or “trending toward significance” is a common way to misinterpret a non-significant p -value. The module detects these sentences so the reader can judge each case in context.

Marginal Significance: You described 3 effects with terms related to ‘marginally significant’ in the Results section. If p values above 0.05 are interpreted as an effect, you inflate the alpha level, and increase the Type 1 error rate. If a p value is higher than the prespecified alpha level, it should be interpreted as a non-significant result.

Table 4.3 shows the p -values that were detected as being interpreted as “marginally significant” across the ten theses in the Results section.

Table 4.3: P -values detected as being interpreted as marginally significant by the `marginal` module

text	section_type
The overall model was almost significant with Wald $\chi^2(12) = 19.01, p = .088$.	results

Gender similarity is marginally significant in predicting collaborated publication patterns ($p = 0.054$). results

Turning back to the results in Table 13, gender is significant in the supervision network ($p = 0.004$) and marginally significant in the publication network ($p = 0.054$), but entirely nonsignificant for the communication network ($p = 0.998$). results

4.4 stat_effect_size module

The `stat_effect_size` module checks whether effect sizes are reported for t -tests and F -tests. Reporting a standardized effect size alongside a test is recommended by the APA Journal Article Reporting Standards, and makes the result more readily usable in meta-analyses.

Running the `stat_effect_size` module on the ten theses, returns the following warning:

Effect Sizes in t-tests and F-tests: We found 26 t-tests and/or F-tests where effect sizes are not reported. Check these tests in the table below, and consider adding effect sizes

Table 4.4 shows the statistical tests detected by the module in Thesis 1, including the extracted test, the effect size and text from which it was extracted. For all four statistical tests, no standardized effect size was reported as indicated by the NA values in the `es` column.

Table 4.4: Statistical tests detected by the `stat_effect_size` module

test_text	es	text
F(31, 62) = 10.37	NA	The test showed that individual differences ($F(31, 62) = 10.37$, $p < .001$) was a significant predictor while XAI condition ($F(2, 62) = 0.08$, $p = .924$) was not.
F(2, 62) = 0.08	NA	The test showed that individual differences ($F(31, 62) = 10.37$, $p < .001$) was a significant predictor while XAI condition ($F(2, 62) = 0.08$, $p = .924$) was not.
F(31, 62) = 8.57	NA	The test showed that both individual differences ($F(31, 62) = 8.57$, $p = .009$) and XAI condition ($F(1, 62) = 7.55$, $p < .001$) are significant predictors.
F(1, 62) = 7.55	NA	The test showed that both individual differences ($F(31, 62) = 8.57$, $p = .009$) and XAI condition ($F(1, 62) = 7.55$, $p < .001$) are significant predictors.

The `stat_effect_size` module also checks whether the reported effect size is coherent with the reported test statistic and degrees of freedom. The module recomputes what the effect size should be from the test statistic and degrees of freedom. It checks Cohen's d and g for t -tests, and η_p^2 and ω_p^2 for F -tests. Table 4.5 shows the first two detected F -tests and the recomputed effect size in Thesis 6. If no standardized effect size is detected, then the `es` column appears as NA.

Table 4.5: Coherence of standardized effect sizes reported along with an F -test by the `stat_effect_size` module

test_text	es	eta_implied_partial	eta_coherence_note
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$F(7, 21) = 5.93$	NA	0.664053751399776	No parseable eta-squared effect size found.
$F(1, 27) = 34.35$	$\eta^2 = .56$	0.559902200488998	Eta-squared reported; cannot be reconstructed from F and dfs in factorial or repeated-measures designs.

4.5 stat_check module

The `stat_check` module recomputes the p -value from each reported test statistic and its degrees of freedom. It flags cases where the reported p -value is inconsistent with the test statistic and degrees of freedom and, more seriously, where the inconsistency changes the statistical significance decision (i.e., a decision error, such as claiming significance when the recomputed p -value is not significant). The `stat_check` module returns results only for t -tests and F -tests and detects only statistics reported in APA format, which is commonly used in psychology journals. Table 4.6 shows the five p -values identified as reporting errors by the module after recomputation.

Table 4.6: Errors in the reporting of p -values detected by `stat_check` module

test_type	reported_p	computed_p	error	decision_error
F	0.009	0.0000000	TRUE	FALSE
F	0.001	0.0078480	TRUE	FALSE
t	0.000	0.0000064	TRUE	FALSE
t	0.023	0.0273977	TRUE	FALSE
t	0.015	0.0191509	TRUE	FALSE

4.6 repo_check module

The `repo_check` module identifies links to research repositories in research outputs including, the Open Science Framework (OSF), GitHub, ResearchBox, PsychArchives, and Zenodo. It then retrieves the list of files in each and reports what has been shared. Table 4.7 displays the summary of the modules, including the number of repositories, files, data files, code files, readme files and naming issues. We can see that two theses reported links to repositories.

Table 4.7: Summary of the `repo_check` module

paper_id	repo_n	files_n	files_data	files_code	files_readme	naming_issues
1	0	0	0	0	0	0
10	0	0	0	0	0	0
2	1	7035	13	13	1	26
3	1	81	47	18	1	26
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0

Table 4.7: Summary of the repo_check module

paper_id	repo_n	files_n	files_data	files_code	files_readme	naming_issues
9	0	0	0	0	0	0

The module also detects naming issues in repository files, such as spaces in file names, which can cause problems when sharing code and data. To illustrate some of the issues that can be detected in file naming, we use Thesis 4, for which the module detected naming issues, as shown in Table 4.7.

Table 4.8 shows the first five naming issues detected by the module in Thesis 4, along with the type of issue and an explanation of each issue. Note that repo_check queries the repository live, so its results reflect the repository's contents at the time the report was rendered rather than at the time it was written: at the time we first checked the repository, it contained 26 naming issues (as reported in Table 4.7 above); since then, the repository has changed, and it now contains 32.

Table 4.8: Naming issues detected by the repo_check module in Thesis 4

file_name	rule	detail
Train test splits.ipynb	spaces	contains a space
Pubmed data.csv	spaces	contains a space
ssc_test (1).csv	spaces	contains a space
ssc_test (1).csv	special-characters	contains a character other than letters, digits, underscore, or dash
3-classes-bert-finetuning (1).ipynb	spaces	contains a space

4.7 power module

The power module uses regular expressions to identify sentences that may describe statistical power analyses and classifies each as a priori, sensitivity, or post-hoc. When collecting data, best practice is to conduct an a priori power analysis to ensure that study design has adequate statistical power to detect the effect size of interest.

Table 4.9 shows the three power analyses detected by the module, including the extracted sentence, the section where it was detected and the type of power analysis assigned. Two could not be classified (i.e., were labelled “unknown”) because they did not contain terms such as “a priori”, “sensitivity” or “post-hoc” while one was classified as “a priori”. It is worth noting that the third power analysis is not a power analysis but rather a statement about how small sample sizes may result in inadequate statistical power.

Table 4.9: Power analyses detected by the power module

pa- per_id	text	power_type
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6	Participants were recruited through a participant database and from the experimenter's network. An a priori power analysis was conducted to determine the required sample. Earlier work by Nazaretsky et al. (2024) examined the effect of feedback source (human vs. AI) on source credibility, reporting a large effect size ($d = 0.88$). To detect an effect size of Cohen's $d = 0.88$, a two-tailed Wilcoxon signed-rank test (matched pairs) was used, with α error = 0.05 and power = 0.9. This resulted in a required sample size of $N = 17$. However, to allow for additional analyses involving multiple within-subject measurements, a repeated-measures ANOVA (within factors) was also considered ($f = 0.40$, α error = 0.05, power = 0.9, two groups, five measurements). This resulted in a required sample size of $N = 28$.	apriori
1	The sample size was calculated based on expected differences in the main dependent variable of interest: self-reported trust. A realistic and meaningful difference between any two conditions would be 0.5 points on a 7-point Likert scale and a 1.53 points standard deviation (Brdnik et al., 2025). Using the Superpower package in R, a sample size of 25 is calculated to achieve a power of 90% (see Appendix A). Five extra participants were added to account for unforeseen issues. Therefore, the recruitment goal for this experiment was 30 participants.	unknown
2	Another notable limitation is the relatively small sample size, which may affect the generalizability of the findings. Currently, we have only collected about 29,000 match-up samples from the Riot API and 5 analyzed video clips for different levels of players with different combinations of melee and ranged champions. Small sample sizes can limit the statistical power of a study and increase the margin of error, potentially leading to less reliable conclusions.	unknown

To assess the completeness of power analysis reporting, we use an LLM to extract and classify information from reported power analysis statements. The LLM is also used to identify potential power analysis statements that do not contain terms such as “a priori”, “sensitivity”, or “post-hoc”. When enabled, the module processes each power analysis statement and extracts structured information, including the statistical test, planned sample size, alpha level, desired power, and effect size. This information is then used to assess whether the power analysis is fully reported.

The LLM functionality can be enabled using `llm_use(TRUE)`. The model is set with `llm_model()`; calling it with no argument returns the current default ("groq"), and passing a model name such as `llm_model("groq/openai/gpt-oss-120b")` changes it. We initially used the small local Ollama model `qwen2.5:3b`, but it misclassified the power analysis in Thesis 1 as post-hoc rather than a priori and failed to detect the reported sample size. We therefore used the larger `openai/gpt-oss-120b` model hosted on [Groq](#). Using Groq requires a free API key, which should be stored as `GROQ_API_KEY` in the `.Renvi` file. For further information on how to run a model locally or through Groq, see the [Metacheck documentation](#).

While testing this with Groq, we ran into a schema bug affecting optional fields (e.g., software,

alpha_level) that are genuinely not mentioned in the source text. Replacing the module's enum-constrained optional fields with a plain nullable string listing the same allowed values in its description avoided the bug; Table 4.10 below uses this workaround. This has been reported upstream to the metacheck maintainers.

Table 4.10 shows the components extracted and classified by the LLM for all three power analysis statements shown in Table 4.9, including the complete column, which represents whether all components of the power analysis were reported. For Thesis 1, the model correctly classified the power analysis as a priori and correctly detected the sample size, desired power, effect size, and software (Superpower), while correctly leaving the statistical test and alpha level missing, since neither was reported in the text. It reported the effect size as a standardized Cohen's d ($0.5 / 1.53 = 0.33$), computed from the raw mean difference and standard deviation given in the text, rather than the unstandardized value of 0.5 as reported by the thesis. For Thesis 6, the model correctly split the paragraph into its two separate power analyses (a Wilcoxon signed-rank test and a repeated-measures ANOVA), correctly extracted both sample sizes, alpha levels, and effect sizes, and correctly left software missing, since none was named in the text. None of the three statements had every field reported, so complete is FALSE throughout — reflecting incompletely reported power analyses in the underlying theses rather than an extraction failure. As with any LLM-based extraction, users should always verify the results before drawing conclusions.

Table 4.10: Classification and completeness of power analyses by the power module using LLM functionality

pa- per_id	power_type	stat_test	n	alpha	power	es	es_met- ric	soft- ware	com- plete
6	apri- ori	Wilcoxon signed- rank test	17	0.05	0.9	0.88	Co- hen's d	NA	FALSE
6	apri- ori	repeated- measures ANOVA	28	0.05	0.9	0.40	Co- hen's f	NA	FALSE
1	apri- ori	NA	25	NA	0.9	NA	NA	Super- power	FALSE

4.8 ref_pubpeer module

The ref_pubpeer module cross-references the cited references against PubPeer, a platform for post-publication peer review. The module reports which cited references have PubPeer comments, how many comments are available, and a link to the corresponding discussion. PubPeer comments do not necessarily indicate that an article is problematic; comments can range from minor requests for clarification to substantive concerns about the research. Identifying cited articles with PubPeer discussions can therefore provide additional information for students and supervisors to consider when evaluating the cited literature.

Table 4.11 shows the cited articles with PubPeer comments detected by the module, along with their corresponding DOIs and links to the PubPeer discussions.

Table 4.11: PubPeer comments on cited articles detected using the ref_pubpeer module

doi	total_comments	url
10.1038/s41579-022-00846-2	1	https://pubpeer.com/publications/51B4F00847F3BDC740ED6B2DA01792
10.3390/app12094429	2	https://pubpeer.com/publications/2A82DBF15C686901A71FD31FA9F880
10.1136/bmj.g7015	1	https://pubpeer.com/publications/BE2197AED4AB91C2338FF0DA99B962
10.1136/bmj.g7015	1	https://pubpeer.com/publications/BE2197AED4AB91C2338FF0DA99B962
10.1136/bmj.g7015	1	https://pubpeer.com/publications/BE2197AED4AB91C2338FF0DA99B962
10.1136/bmj.g7015	1	https://pubpeer.com/publications/BE2197AED4AB91C2338FF0DA99B962

5 Summary

Overall, these examples illustrate how Metacheck can be used to identify potential issues in master's theses from both the student's perspective, as a tool for identifying and addressing common issues before submission, and the supervisor's perspective, as a first layer of quality control that can help identify areas requiring further attention. However, automated tools such as Metacheck are not infallible and may produce both false positives and false negatives. The results should therefore be interpreted as indicators of potential issues rather than definitive judgments. Ultimately, it remains the responsibility of the student or supervisor to review the flagged issues, assess their relevance in context, and determine whether any action is required.

6 Appendix A

Survey on Automated Feedback for Master Theses

Dear participant,

We are developing new modules for a tool designed to help master students and their supervisors check for common mistakes and improve adherence to best research practices in theses. This tool currently detects issues such as missing sample size justifications, incorrect statistical interpretations of non-significant p -values, and references to retracted papers.

This survey aims to identify common feedback provided by supervisors on master's theses, focusing on recurring mistakes related to structural aspects, research methods and statistical reporting, and citations. Additionally, we also seek to understand which aspects of feedback supervisors believe could be automated to better assist students in the process.

Examples of these automated checks include:

- verifying the presence/absence of section headers;
- analyzing the number of references and their citation frequency;
- and checking whether tables and figures are referenced in the main text.

To develop these new modules, we are exploring different approaches, including rule-based text searches (e.g., identifying specific phrases indicating common mistakes) and using large language models (LLMs) to detect issues and provide automated suggestions. Note that we only use LLM and API providers with high data security standards, ensuring that any data processed by the tool will not be shared with third parties or used for model training.

Your input will help shape the development of this tool, ensuring that it addresses real challenges faced by students and supervisors. The survey should take approximately **5 minutes** to complete.

Thank you for your time and insights!

Q1: What are some of the most common types of feedback you provide to master students on their theses? Please focus on recurring issues related to the structure of the master thesis (e.g., missing headings), research methods (e.g., not reporting statistical tests, poor reporting practices (not reporting exact p -values), citations (missing reference), or other structural aspects.

Q2: Are there any aspects of the feedback process that you think could be (partially) automated?

Q3: Are there any aspects of thesis feedback that you believe should not be automated?

Q4: Do you or your students follow any field-specific reporting guidelines when writing/critiquing the thesis? For instance, social scientists and psychologists often use the American Psychological Association (APA) guidelines to format manuscripts and report results.

Q5: Is there anything else you would like to share with us?